

Three Economic Models, One Energy Mix

The same US wind + PV + solar-thermal + CSP + sand-battery system, simulated under capitalist markets, RCED democratic planning, and Soviet central planning

Same sun, same wind, same hardware. The climate and resource outcome is decided by the institutions, not the technology — and each of the three systems fails, or succeeds, on a different axis.

Executive Summary

This study applies a single physical energy system — a fully electrified United States running on solar PV, wind, concentrated solar power, solar thermal, a seasonal sand battery and residential plus utility batteries — to three different economic frameworks. The demand for energy *services* (homes heated, industry run, vehicles moved) is held identical throughout, so that only the institutions vary. The capitalist market is the anchor case; it is then compared with the RCED's democratic planning and with Soviet-style central planning.

The three systems separate cleanly along two axes — carbon and resource use. The **market** is capital-efficient (it builds the least hardware) but leaves the climate problem largely unsolved, emitting about **1011 Mt CO₂/yr** because it will not finance seasonal storage or heat networks. **Soviet** planning decarbonizes almost completely (~20 Mt) but at enormous waste: it delivers **+48%** more energy and builds **~8550 GW** of capacity to provide the same services. Only the **RCED** reaches the ideal corner — near-zero carbon *and* low resource use — by combining the market's efficiency with the plan's ability to build non-marketable infrastructure.

1. The three economic models

1.1 Capitalist market (current US).

Investment follows profitability and hourly prices. Short-duration batteries (and residential storage) are built because daily arbitrage pays; seasonal storage is not, because an asset that cycles once a year cannot recover its capital. Urban heat networks barely exist, so the sand battery cannot reach buildings. Natural gas remains the flexible resource, and the merchant fleet stops growing once curtailment makes the marginal renewable unprofitable.

1.2 RCED democratic planning.

A citizens' assembly allocates investment, and ownership is by use rather than by capital. The plan builds exactly what the market will not: inter-seasonal sand storage, urban heat networks, a national grid, and a deliberate, modest overbuild. Crucially it also disciplines demand through deep building renovation and use-based property, so the system is efficient as well as planned.

1.3 Soviet central planning.

A central authority can mobilize capital for heavy infrastructure — it builds the seasonal storage, the national grid and, historically a Soviet strength, extensive district heating. But near-free energy and gross-output targets (the "val") remove any incentive to conserve: buildings are poorly insulated, heat is unregulated, industry is energy-intensive, and the plan over-invests to inflate tonnage. Supply is reliable, but through brute-force overbuild and pervasive waste.

2. Results

Indicator	Market (capitalist)	RCED (democratic)	USSR (command)
Reliability (power / heat)	100% (gas-backed)	100%	100% / 99.5%
Residual CO2 (Mt/yr)	1011	~5	20
Renewable share of electricity	84%	100%	100%
Heat supplied by fossil gas	70%	0%	1%
Installed renewable capacity (GW)	2,612	4,750	8550
Final energy for same services (TWh)	~11,400	11,400	16837 (+48%)

USA — ONE RENEWABLE MIX UNDER THREE ECONOMIC SYSTEMS Each system fails (or succeeds) on a different axis

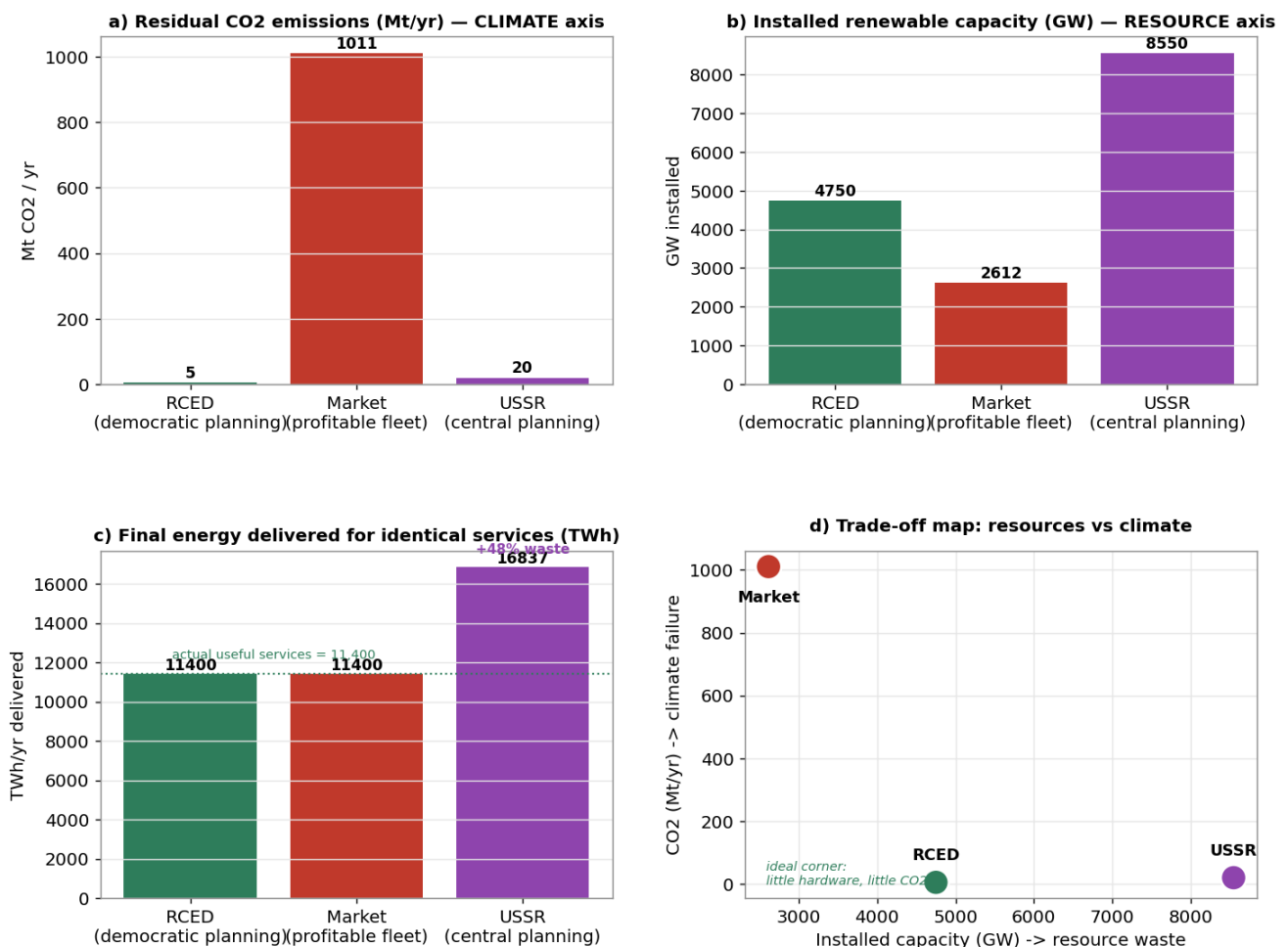


Figure 1 — The three systems separate on two axes. (a) carbon; (b) installed hardware; (c) energy delivered for identical services; (d) the trade-off map, with RCED alone in the ideal low-hardware/low-carbon corner.

3. Analysis

3.1 Each system fails on a different axis.

The trade-off map (Figure 1d) is the heart of the result. The market sits top-left: little hardware, but ~1 Gt of CO2 — it under-invests in the seasonal layer and stays tied to gas. The USSR sits bottom-right: low carbon, but a mountain of capacity and wasted energy — decarbonization by brute force. The RCED occupies the bottom-left ideal corner: low carbon and lean, because it pairs the plan's ability to build

seasonal storage and heat networks with market-like efficiency and genuine demand discipline.

3.2 Why the market case cannot be rescued by batteries.

A natural objection to the market result is: why not just add residential batteries? A dedicated sweep shows the limit. In the market case, adding home batteries up to every US household (~1.5 TWh) cuts emissions by only single-digit-to-low-double-digit percent, and the curve plateaus thereafter. Short-duration batteries shift midday solar to the evening — the daily cycle — but they are empty during multi-day winter lulls and, decisively, they cannot store or deliver heat. With no heat networks, roughly three-quarters of the residual emissions are heat that batteries categorically cannot touch. Batteries are the right tool for the day, not for the season.

3.3 Residential batteries in a complete system.

The same residential batteries become genuinely useful once the rest of the infrastructure exists. In the full RCED-style system, a fleet covering every US home cycles about 150 times a year, doing the entire daily electricity job, while the sand battery cycles roughly once a year, moving heat from summer to winter. The two are complementary, each covering a timescale the other cannot. The lesson is not that home batteries are useless, but that they only pay off when seasonal storage and heat infrastructure are also in place — which is precisely what the market does not build.

4. Limitations

- Demand for services is held identical to isolate the institutional effect; the Soviet mix is also held renewable for comparability, whereas a historical Soviet plan would have centralized on coal and nuclear, giving higher emissions. The ~20 Mt figure is thus a favorable floor.
- The market merchant fleet is sized by a curtailment heuristic, not a full price-based investment model; gas is treated as unlimited flexibility.
- Soviet inefficiency (+48% demand, x1.8 capacity) is calibrated on documented Soviet energy intensity but remains an order-of-magnitude estimate.
- Synthetic weather, a single representative year, and a single national node (no inter-regional congestion).

5. Conclusion

The same renewable hardware can leave the United States emitting on the order of a gigatonne of CO₂ a year, or decarbonize it almost completely — and it can do the latter either leanly or with vast waste. Which outcome occurs is set by the economic system, not the technology. The market is efficient with capital but will not finance the seasonal storage and heat networks that winter demands, so it stalls on the hard tranche and leans on gas. Soviet-style planning will build that infrastructure and reach near-zero carbon, but its lack of demand discipline and its gross-output incentives squander energy and materials on a massive scale.

The RCED's claim, tested here, is that a third path exists: democratic planning that builds the non-marketable infrastructure like the command economy, but couples it with efficiency and demand discipline like a well-functioning market, and with use-based ownership that removes the rent motive driving both waste and under-investment. In this simulation that combination is the only one to reach the corner that matters — deep decarbonization at low resource cost. Whether such an institution is politically achievable is beyond the model; what the model shows is that, physically and economically, the binding constraint on the energy transition is institutional design far more than engineering.